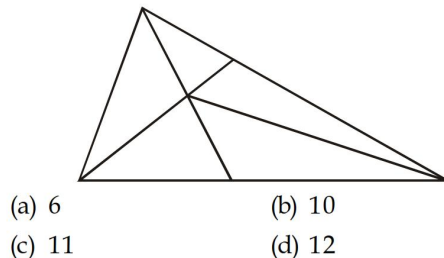


Figure Counting

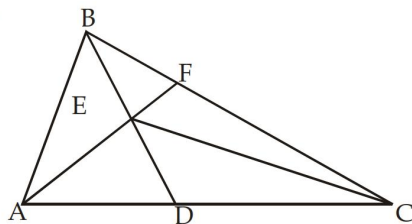
It involves the problems related to the counting the number of geometrical figures in a given complex figure. The systematic method for determining the number of any particular type of figure by the analysis of the complex figure would be clear from the examples that are given below: -

Solved Examples

1. How many triangles are there in the following figure?

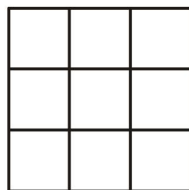


Sol.



Δ 's, ABE, BEF, EFC, CDE, AED
ABF, BCE, ACE, ABD,
AFC, BCD, & ABC
the answer is (d), 12

2. Count the number of squares



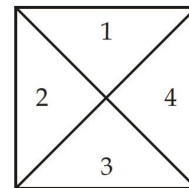
- (a) 18 (b) 14
(c) 10 (d) 9

Sol. If the number of square are equal in horizontal and vertical direction, then we use formula

$$N = n^2 + (n-1)^2 + (n-2)^2 + \dots$$

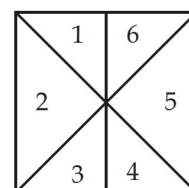
$$\therefore \text{No of square} = 3^2 + 2^2 + 1^2 = 9 + 4 + 1 = 14$$

3. Count the number of triangle in the given square.



Sol. No. of triangles = $4 \times 2 = 8$

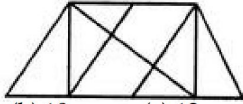
4. Count the no. of triangle in a given square.



Sol. No. of triangles = $6 \times 2 = 12$

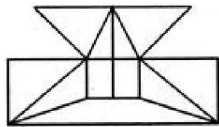
Practice Set

1. Find the number of triangles in the given figure.



(a) 8 (b) 10 (c) 12 (d) 14

2. Find the minimum number of straight lines required to make the given figure.



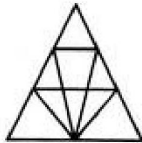
(a) 16 (b) 17 (c) 18 (d) 19

3. Find the number of triangles in the given figure.



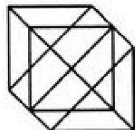
(a) 22 (b) 24 (c) 26 (d) 28

4. Find the number of triangles in the given figure.



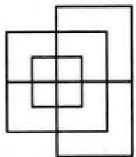
(a) 12 (b) 18 (c) 22 (d) 26

5. Find the number of triangles in the given figure.



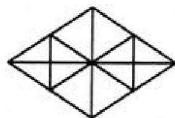
(a) 18 (b) 20 (c) 24 (d) 27

6. Find the minimum number of straight lines required to make the given figure.



(a) 13 (b) 15 (c) 17 (d) 19

7. Find the number of triangles in the given figure.



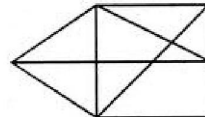
(a) 16 (b) 22 (c) 28 (d) 32

8. Find the number of triangles in the given figure.



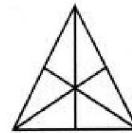
(a) 11 (b) 13 (c) 15 (d) 17

9. Find the number of triangles in the given figure.



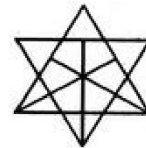
(a) 12 (b) 13 (c) 14 (d) 15

10. Find the number of triangles in the given figure.



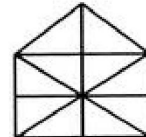
(a) 16 (b) 13 (c) 9 (d) 7

11. Find the number of triangles in the given figure.



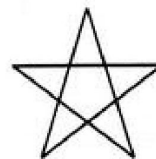
(a) 21 (b) 23 (c) 25 (d) 27

12. Find the number of triangles in the given figure.



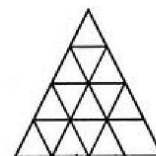
(a) 18 (b) 20 (c) 21 (d) 25

13. Find the number of triangles in the given figure.



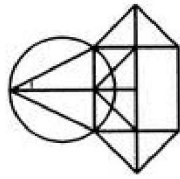
(a) 5 (b) 9 (c) 12 (d) 10

14. Find the minimum number of straight lines required to make the given figure.

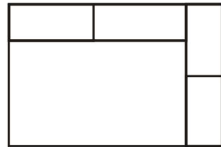


(a) 9 (b) 11 (c) 15 (d) 16

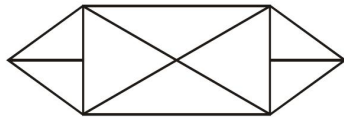
15. Find the number of triangles in the given figure.



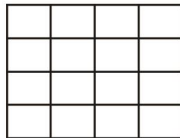
- (a) 10 (b) 12 (c) 14 (d) 16
16. How many rectangles are there in the given figure ?



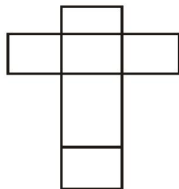
- (a) 6 (b) 7 (c) 8 (d) 9
17. How many triangles are there in the following figure?



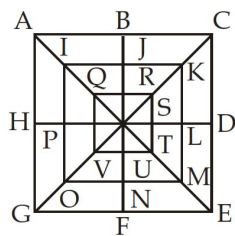
- (a) 12 (b) 13 (c) 14 (d) 15
18. How many squares are there in the given figure?



- (a) 35 (b) 40 (c) 50 (d) 30
19. How many rectangles are there in the given figure?



- (a) 12 (b) 15 (c) 17 (d) 18
20. What is the number of straight lines in the following figure ?

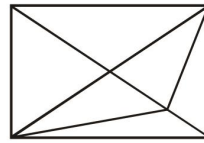


- (a) 10 (b) 15 (c) 16 (d) 18

Distinct Questions

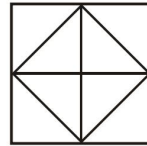
Direction (21 - 28) : In each of the following question, find the number of triangles in the given figure.

21.



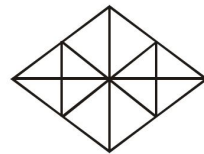
- (a) 11 (b) 13 (c) 15 (d) 17

22.



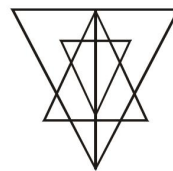
- (a) 8 (b) 10 (c) 12 (d) 14

23.



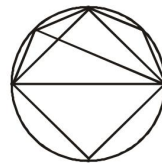
- (a) 16 (b) 22 (c) 28 (d) 32

24.



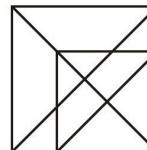
- (a) 27 (b) 25 (c) 23 (d) 21

25.



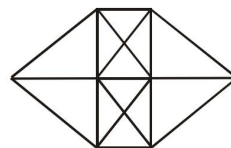
- (a) 8 (b) 10 (c) 11 (d) 12

26.



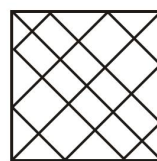
- (a) 16 (b) 18 (c) 19 (d) 21

27.



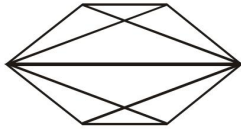
- (a) 20 (b) 24 (c) 28 (d) 32

28.



- (a) 28 (b) 32 (c) 36 (d) 40

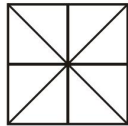
Direction (29 – 30):



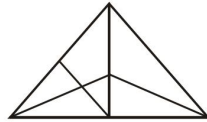
29. Find the number quadrilaterals in the given figure.
 (a) 6 (b) 7 (c) 9 (d) 11
 30. What is the number of pentagons in the given figure?
 (a) 2 (b) 3 (c) 4 (d) 6

Previous year Questions (Memory Based)

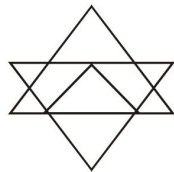
1. How many triangles are there in the following figure? 7. In figure A B C D E F how many triangles are there?



- (a) 16 (b) 14 (c) 8 (d) 12
 2. How many triangles are there in the following figure? 8. How many triangles are there in the following figure?



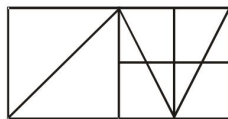
- (a) 32 (b) 13 (c) 24 (d) 48
 3. How many triangles are there in the following figure?



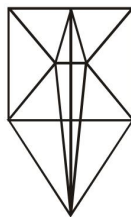
- (a) 10 (b) 11 (c) 12 (d) 13
 4. How many triangles are there in the following figures?



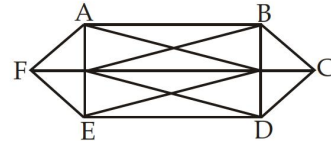
- (a) 15 (b) 16 (c) 17 (d) 18
 5. How many triangles are there in the following figure?



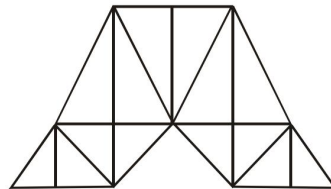
- (a) 13 (b) 14 (c) 15 (d) 16
 6. How many triangles are there in the following figure?



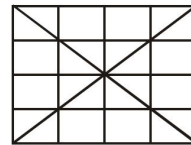
- (a) 45 (b) 24 (c) 28 (d) 20



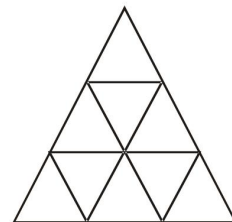
- (a) 24 (b) 26 (c) 28 (d) 30
 8. How many triangles are there in the following figure?



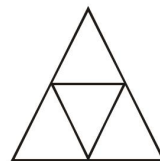
- (a) 27 (b) 23 (c) 29 (d) 36
 9. How many triangles are there in the following figure?



- (a) 32 (b) 16 (c) 24 (d) 48
 10. How many triangles are there in the following figure?

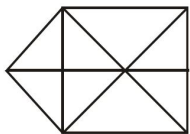


- (a) 13 (b) 15 (c) 17 (d) 20
 (11 – 15):
 11. Find the number of triangles in the given figure.



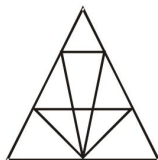
- (a) 4 (b) 5 (c) 6 (d) 7

12.



- (a) 15 (b) 16 (c) 17 (d) 18

13.



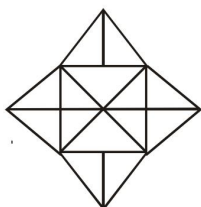
- (a) 12 (b) 18 (c) 22 (d) 26

14.



- (a) 16 (b) 18 (c) 14 (d) 15

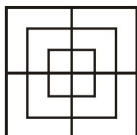
15.



- (a) 18 (b) 20 (c) 28 (d) 34

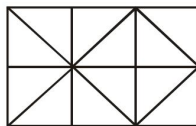
(16 - 17): In each of the following questions, count the number of squares in the given figure.

16.



- (a) 8 (b) 12 (c) 15 (d) 18

17.



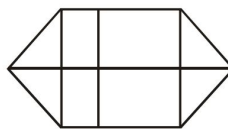
- (a) 6 (b) 7 (c) 9 (d) 10

18. How many triangles and parallelograms are there in the following figure?



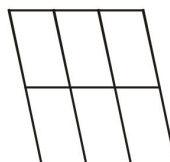
- (a) 21, 17 (b) 19, 13 (c) 21, 15 (d) 19, 17

19. How many rectangles are there in the following figure?



- (a) 10 (b) 9 (c) 8 (d) 7

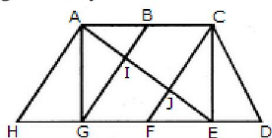
20. Count the number of parallelograms in the given figure.



- (a) 20 (b) 18 (c) 16 (d) 12

Practice Set Solutions

1. (d); The figure may be labeled as shown.



The simplest triangles are AHG, AIG, AIB, JFE, CJE and CED i.e. 6 in number.

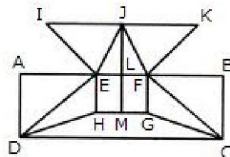
The triangles composed of two components each are ABG, CFE, ACJ and EGI i.e. 4 in number.

The triangles composed of three components each are ACE, AGE and CFD i.e. 3 in number.

There is only one triangle i.e. AHE composed of four components.

Therefore, there are $6 + 4 + 3 + 1 = 14$ triangles in the given figure.

2. (b); The figure may be labeled as shown.



The horizontal lines are IK, AB, HG and DC i.e. 4 in number.

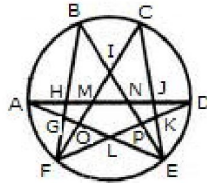
The vertical lines are AD, EH, JM, FG and BC i.e. 5 in number.

The slanting lines are IE, JE, JF, KF, DE, DH, FC

and GC i.e. 8 is number.

Thus, there are $4 + 5 + 8 = 17$ straight lines in the figure.

3. (d); The figure may be labeled as shown.



The simplest triangles are AGH, GFO, LFO, DJK, EKP, PEL and IMN i.e. 7 in number.

The triangles having two components each are GFL, KEL, AMO, NDP, BHN, CMJ, NEJ and HFM i.e. 8 in number.

The triangles having three components each are IOE, IFP, BIF and CEI i.e. 4 in number.

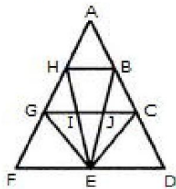
The triangles having four components each are ANE and DMF i.e. 2 in number.

The triangles having five components each are FCK, BGE and ADL i.e. 3 in number.

The triangles having six components each are BPF, COE, DHF and AJE i.e. 4 in number.

Total number of triangles in the figure = $7 + 8 + 4 + 2 + 3 + 4 = 28$.

4. (b); The figure may be labeled as shown.



The simplest triangles are AHB, GHI, BJC, GFE, GIE, IJE, CEJ and CDE i.e. 8 in number.

The triangles composed of two components each are HEG, BEC, HBE, JGE and ICE i.e. 5 in number.

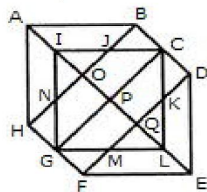
The triangles composed of three components each are FHE, GCE and BED i.e. 3 in number.

There is only one triangle i.e. AGC composed of four components.

There is only one triangle i.e. AFD composed of nine components.

Thus, there are $8 + 5 + 3 + 1 + 1 = 18$ triangles in the given figure.

5. (c); The figure may be labeled as shown.



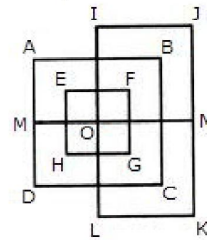
The simplest triangles are IJO, BCJ, CDK, KQL, MLQ, GFM, GHN and NIO i.e. 8 in number.

The triangles composed of two components each are ABO, AHO, NIJ, IGP, ICP, DEQ, FEQ, KLM, LCP and LGP i.e. 10 in number.

The triangles composed of four components each are HAB, DEF, LGI, GIC, ICL and GLC i.e. 6 in number.

Total number of triangles in the figure = $8 + 10 + 6 = 24$.

6. (a); The figure may be labeled as shown.

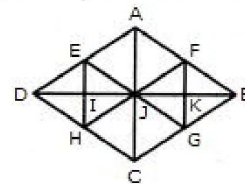


The horizontal lines are IJ, AB, EF, MN, HG, DC and LK i.e. 7 in number.

The vertical lines are AD, EH, IL, FG, BC and JK i.e. 6 in number.

Thus, there are $7 + 6 = 13$ straight lines in the figure.

7. (c); The figure may be labeled as shown.



The simplest triangles are AFJ, FJK, FKB, BKG, JKG, JGC, HJC, HIJ, DIH, DEI, EIJ and AEJ i.e. 12 in number.

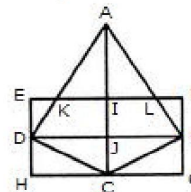
The triangles composed of two components each are JFB, FBG, BJG, JFG, DEJ, EJH, DJH and DEH i.e. 8 in number.

The triangles composed of three components each are AJB, JBC, DJC and ADJ i.e. 4 in number.

The triangles composed of six components each are DAB, ABC, BCD and ADC i.e. 4 in number.

Thus, there are $12 + 8 + 4 + 4 = 28$ triangles in the figure.

8. (c); The figure may be labeled as shown.



The simplest triangles are AKI, AIL, EKD, LFB, DJC, BJC, DHC and BCG i.e. 8 in number.

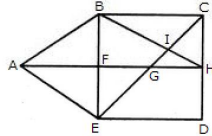
The triangles composed of two components each are AKL, ADJ, AJB and DBC i.e. 4 in number.

The triangles composed of the three components each are ADC and ABC i.e. 2 in number.

There is only one triangle i.e. ADB composed of four components.

Thus, there are $8 + 4 + 2 + 1 = 15$ triangles in the figure.

9. (d); The figure may be labeled as shown.



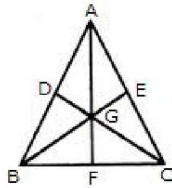
The simplest triangles are ABF, BIC, CIH, GIH, FGE and AFE i.e. 6 in number.

The triangles composed of two components each are ABE, AGE, BHF, BCH, CGH and BIE i.e. 6 in number.

The triangles composed of three components each are ABH, BCE and CDE i.e. 3 in number.

Hence, the total number of triangles in the figure = $6 + 6 + 3 = 15$.

10. (a); The figure may be labeled as shown.



The simplest triangles are AGE, EGC, GFC, BGF, DGB and ADG i.e. 6 in number.

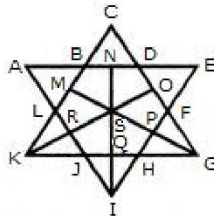
The triangles composed of two components each are AGC, BGC and ABG i.e. 3 in number.

The triangles composed of three components each are AFC, BEC, BDC, ABF, ABE and DAC i.e. 6 in number.

There is only one triangle i.e. ABC composed of six components.

Thus, there are $6 + 3 + 6 + 1 = 16$ triangles in the given figure.

11. (d); The figure may be labeled as shown.



The simplest triangles are ABL, BCD, DEF, FGP, PGH, QHI, JQI, KRJ and LRK i.e. 9 in number.

The triangles composed of two components each are OSG, SGQ, SPI, SRI, KSQ, KMS, FGH, JHI and JKL i.e. 9 in number.

There is only one triangle i.e. KSG which is composed of four components.

The triangles composed of five components each are NEI, ANI, MCG and KCO i.e. 4 in number.

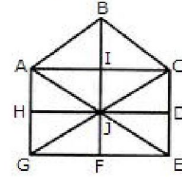
The triangles composed of six components each are GMK and KOG i.e. 2 in number.

There is only one triangle i.e. AEI composed of ten components.

There is only one triangle i.e. KCG composed of eleven components.

Therefore, Total number of triangles in the given figure = $9 + 9 + 1 + 4 + 2 + 1 + 1 = 27$.

12. (c); The figure may be labeled as shown.



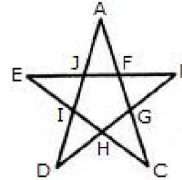
The simplest triangles are ABI, BIC, AIJ, CIJ, AHJ, CDJ, JHG, JDE, GJF and EJJ i.e. 10 in number.

The triangles composed of two components each are ABC, BCJ, ACJ, BAJ, AJG, CJE and GJE i.e. 7 in number.

The triangles composed of four components each are ACG, ACE, CGE and AGE i.e. 4 in number.

Total number of triangles in the figure = $10 + 7 + 4 = 21$.

13. (d); The figure may be labeled as shown.

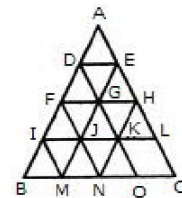


The simplest triangles are AJF, FBG, GCH, HDI and IEJ i.e. 5 in number.

The triangles composed of three components each are EBH, AIC, EFC, ADG and BJD i.e. 5 in number.

Thus, there are $5 + 5 = 10$ triangles in the figure.

14. (b); The figure may be labeled as shown.

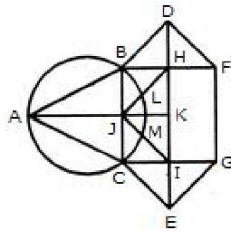


The horizontal lines are DE, FH, IL and BC i.e. 4 in number.

The slanting lines are AC, DO, FN, IM, AB, EM and HN i.e. 7 in number.

Thus, there are $4 + 7 = 11$ straight lines in the figure.

15. (c); The figure may be labeled as shown.

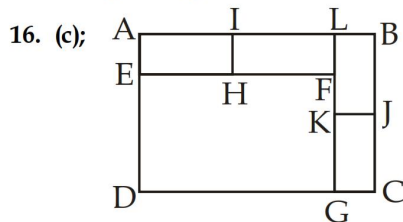


The simplest triangles are ABJ, ACJ, BDH, DHF, CIE and GIE i.e. 6 in number.

The triangles composed of two components each are ABC, BDF, CEG, BHJ, JHK, JKI and CJI i.e. 7 in number.

There is only one triangle JHI which is composed of four components.

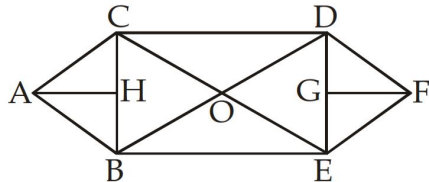
Thus, there are $6 + 7 + 1 = 14$ triangles in the given figure.



Required rectangles :-

AIHE, ILFH, ALFE, LBJK, JCGK, LBCG, EFGD, ABCD

17. (c); $\rightarrow 14$

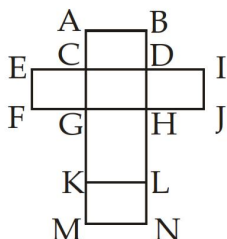


Δ 's ABC, ACH, ABH, DEF, DGF, GEF
BDE, CDE, BEC, BOE, EOD, DOC
BOC, BDC

18. (d); $\rightarrow 30$

$$4^2 + 3^2 + 2^2 + 1^2 = 16 + 9 + 4 + 1 = 30$$

19. (b); $\rightarrow 15$



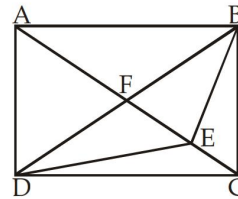
ABCD, ECFG, GCDH, HDIJ, GKLH, KMN, L,
EFJL, EFHD, GCIJ, ABNM, ABLK, MNDC,
ABGH, GHNM, CDLK

20. (c); $\rightarrow 16$

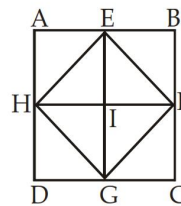
Straight lines : - AC, CE, EG, AG, IK, IO, OM, MK,
QS, ST, TV, VQ, BF, HD, GC, AE

Distinct Solutions

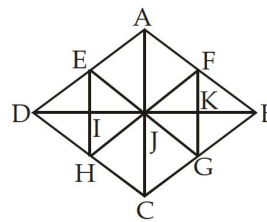
21. (c); AFB, FEB, EBC, DEC, DFE, AFD $\rightarrow 6 \Delta$'s
AEB, FBC, DFC, ADE, DBE, ABD $\rightarrow 6 \Delta$'s
ADC and ABC $\rightarrow 2 \Delta$'s
DBC $\rightarrow 1 \Delta$
Total = $6 + 6 + 2 + 1 = 15 \Delta$'s



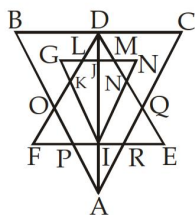
22. (c); AEH, EBF, EFI, FGC, EHI, IFG, DGH, HIG $\rightarrow 8 \Delta$'s
HEF, EFG, HFG, EHG $\rightarrow 4 \Delta$'s
Total = $8 + 4 = 12$ triangles in the figure.



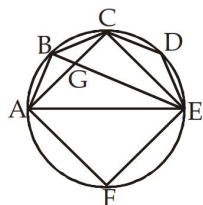
23. (c); AFJ, FJK, FKB, BKG, JKG, JGC, HJC, HIJ, DIH,
DEI, EIJ, AEJ $\rightarrow 12 \Delta$'s
JFB, FBG, BJG, JFG, DEJ, EJH, DJH, DEH $\rightarrow 8 \Delta$'s
AJB, JBC, DJC, ADJ $\rightarrow 4 \Delta$'s
DAB, ABC, BCD, ADC $\rightarrow 4 \Delta$'s
Total = $12 + 8 + 4 + 4 = 28 \Delta$'s



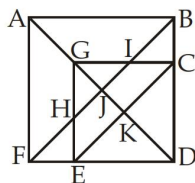
24. (a); GLK, DLJ, DJM, HMN, QRE, IRA, IPA, FPO $\rightarrow 8 \Delta$'s
BDO, CDQ, DLM, PRA, KFI, NEI, HJI, GJI, DKI
and DNI $\rightarrow 10 \Delta$'s
DIE, DFI, DOA, DQA and GHI $\rightarrow 5 \Delta$'s
DCA and DBA $\rightarrow 2 \Delta$'s
DEF $\rightarrow 1 \Delta$'s
ABC $\rightarrow 1 \Delta$'s
Total = $8 + 10 + 5 + 2 + 1 + 1 = 27 \Delta$'s



25. (b); $\triangle ABG, \triangle BCG, \triangle CGE, \triangle CDE, \triangle AGE$ and $\triangle AEF \rightarrow 6 \triangle$'s
 $\triangle ABE, \triangle ABC, \triangle BCE$ and $\triangle ACE \rightarrow 4 \triangle$'s
 $\therefore \text{Total} = 6 + 4 = 10 \triangle$'s

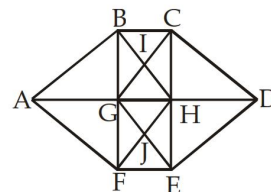


26. (d); $\triangle EFH, \triangle BIC, \triangle GHJ, \triangle GIJ, \triangle EKD$ and $\triangle CKD \rightarrow 6 \triangle$'s
 $\triangle ABJ, \triangle AFJ, \triangle GCK, \triangle GEK, \triangle CED, \triangle GIH \rightarrow 6 \triangle$'s
 $\triangle GCD, \triangle GED, \triangle DJB, \triangle DJF \rightarrow 4 \triangle$'s
 $\triangle ABF$ and $\triangle GCE \rightarrow 2 \triangle$'s
 $\triangle ABD$ and $\triangle AFD \rightarrow 2 \triangle$'s
 $\triangle FBD \rightarrow 1 \triangle$
 $\therefore \text{Total} = 6 + 6 + 4 + 2 + 2 + = 21 \triangle$'s

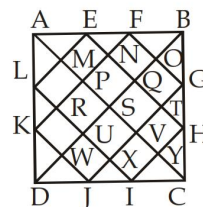


27. (c); $\triangle ABG, \triangle BIG, \triangle BIC, \triangle CIH, \triangle GIH, \triangle CDH, \triangle HED, \triangle GHJ, \triangle HJE, \triangle FEJ, \triangle GFJ$ and $\triangle AGF \rightarrow 12 \triangle$'s
 $\triangle ABF, \triangle CDE, \triangle GBC, \triangle BCH, \triangle CHG, \triangle BHG, \triangle GHF, \triangle GHE, \triangle HEF$ and $\triangle GEF \rightarrow 10 \triangle$'s
 $\triangle ABH, \triangle AFH, \triangle CDG$ and $\triangle GDE \rightarrow 4 \triangle$'s
 $\triangle BHF, \triangle CGE \rightarrow 2 \triangle$'s

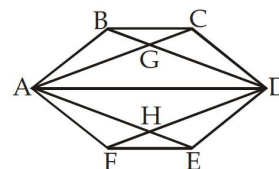
Total = $12 + 10 + 4 + 2 = 28 \triangle$'s



28. (c); $\triangle AML, \triangle LRK, \triangle KWD, \triangle DWJ, \triangle JXI, \triangle IYC, \triangle CYH, \triangle HTG, \triangle GOB, \triangle BOF, \triangle FNE$ and $\triangle EMA \rightarrow 12 \triangle$'s
 $\triangle AEL, \triangle KDJ, \triangle HIC$ and $\triangle FBG \rightarrow 4 \triangle$'s
 $\triangle APF, \triangle EQB, \triangle BQH, \triangle GVC, \triangle CVJ, \triangle IUD, \triangle DUL$ and $\triangle KPA \rightarrow 8 \triangle$'s
 $\triangle ASB, \triangle BSC, \triangle CSD, \triangle DSA, \triangle AKF, \triangle EBH, \triangle GCJ, \triangle IDL \rightarrow 8 \triangle$'s
 $\triangle ADB, \triangle ABC, \triangle BCD$ and $\triangle CDA \rightarrow 4 \triangle$'s
 $\therefore \text{Total} = 12 + 4 + 8 + 8 + 4 = 36 \triangle$'s



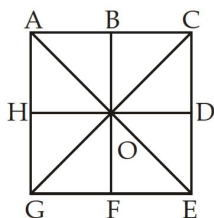
29. (d); $\triangle ABCD, \triangle ABDE, \triangle ABDF, \triangle ABDH, \triangle CDHA, \triangle CDEA, \triangle CDFA, \triangle DEAG, \triangle DEFA, \triangle FAGD$ and $\triangle AGDH$.
 $\therefore$ The number of quadrilaterals in the figure is 11.



30. (d); The pentagons in the figure are $\triangle ABCDE, \triangle ABCDF, \triangle ABCDH, \triangle ABDEF, \triangle CDEFA$ and $\triangle DEFAG$.
 Clearly, there are six pentagons in the figure.

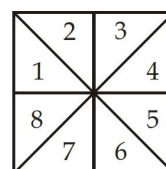
Previous year Solutions (Memory Based)

1. (a);



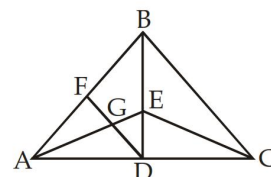
$\triangle GEC, \triangle GAC, \triangle ECA, \triangle EGA, \triangle GOE$
 $\triangle COE, \triangle AOC, \triangle AOG, \triangle GOF, \triangle FOE$
 $\triangle EOD, \triangle DOC, \triangle BOC, \triangle AOB, \triangle AOH$
 $\triangle HOG$

or



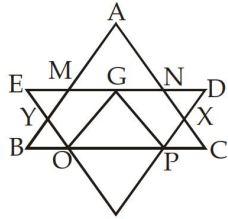
No. of triangles = $8 \times 2 = 16$

2. (b);



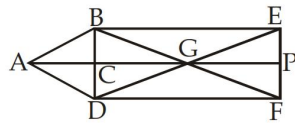
$\triangle ABC$ $\triangle ABD$ $\triangle BDC$ $\triangle ACE$ $\triangle AGD$ $\triangle AGF$
 $\triangle DEG$ $\triangle ABE$ $\triangle BEC$ $\triangle DEC$ $\triangle DFB$ $\triangle AFD$
 $\triangle AED$

3. (b);



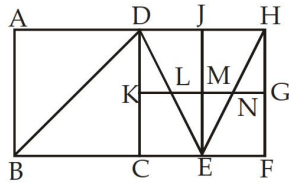
$\triangle ABC$ $\triangle EFD$ $\triangle GOP$ $\triangle MAN$ $\triangle EMY$ $\triangle BOY$
 $\triangle FOP$ $\triangle PXC$ $\triangle NXD$ $\triangle OGE$ $\triangle GDP$

4. (c);



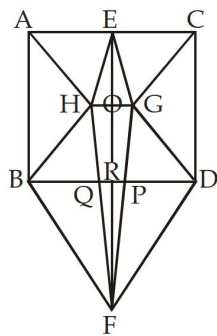
$\triangle ABD$ $\triangle ABC$ $\triangle ACD$ $\triangle BEF$ $\triangle BDF$ $\triangle DEF$
 $\triangle DEB$ $\triangle DGF$ $\triangle GFE$ $\triangle BEG$ $\triangle BGD$ $\triangle BCG$
 $\triangle DCG$ $\triangle EGP$ $\triangle PGF$ $\triangle ABG$ $\triangle ADG$

5. (a);



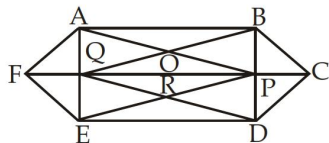
$\triangle ADB$ $\triangle BDC$ $\triangle BDE$ $\triangle DEH$ $\triangle DCE$ $\triangle DEJ$
 $\triangle EJH$ $\triangle EHF$ $\triangle HNG$ $\triangle KDL$ $\triangle ELN$ $\triangle LEM$
 $\triangle MNE$

6. (c);



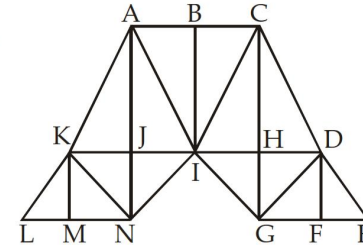
$\triangle ABH$ $\triangle DGC$ $\triangle HGE$ $\triangle HOE$ $\triangle OGE$ $\triangle HFG$
 $\triangle HOF$ $\triangle GOF$ $\triangle AHE$ $\triangle EGC$ $\triangle BQH$ $\triangle PGD$
 $\triangle BFD$ $\triangle BRF$ $\triangle DRF$ $\triangle DPF$ $\triangle PFB$ $\triangle FQB$
 $\triangle FQD$ $\triangle FGD$ $\triangle BHF$ $\triangle BED$ $\triangle BER$ $\triangle DRE$
 $\triangle AEB$ $\triangle DEC$ $\triangle EFH$ $\triangle EFG$

7. (c);



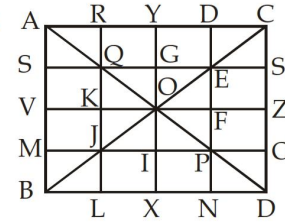
$\triangle AFE$ $\triangle AQF$ $\triangle FQE$ $\triangle BDC$ $\triangle BPC$ $\triangle PDC$
 $\triangle AQB$ $\triangle PQB$ $\triangle ABP$ $\triangle APQ$ $\triangle AOB$ $\triangle BOP$
 $\triangle POQ$ $\triangle AOQ$ $\triangle DPQ$ $\triangle DQE$ $\triangle EPD$ $\triangle DQP$
 $\triangle ERD$ $\triangle DRP$ $\triangle PRQ$ $\triangle QRE$ $\triangle AFP$ $\triangle FPE$
 $\triangle BQC$ $\triangle DQC$ $\triangle AEP$ $\triangle BDQ$

8. (c);



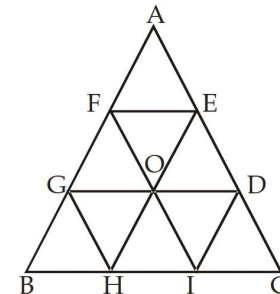
$\triangle AKI$ $\triangle AKJ$ $\triangle AJI$ $\triangle AIC$ $\triangle ABI$ $\triangle BCI$
 $\triangle CID$ $\triangle CIH$ $\triangle CHD$ $\triangle KLN$ $\triangle KLM$ $\triangle KMN$
 $\triangle KNI$ $\triangle KJN$ $\triangle JNI$ $\triangle JGD$ $\triangle IHG$ $\triangle GHG$
 $\triangle GDE$ $\triangle GFD$ $\triangle FDE$ $\triangle ANK$ $\triangle ANI$ $\triangle CGI$
 $\triangle CGD$ $\triangle ACN$ $\triangle ACG$ $\triangle ALN$ $\triangle CGE$

9. (d);



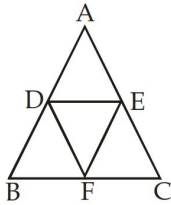
$\triangle ABC$ $\triangle BCD$ $\triangle ACD$ $\triangle ADB$ $\triangle AOB$ $\triangle BOD$
 $\triangle DOC$ $\triangle AOC$ $\triangle CSE$ $\triangle ECD$ $\triangle OEF$ $\triangle OEG$
 $\triangle JIO$ $\triangle KOJ$ $\triangle JBL$ $\triangle JBM$ $\triangle AQR$ $\triangle ASQ$
 $\triangle QGO$ $\triangle QOK$ $\triangle OIP$ $\triangle OPF$ $\triangle NPD$ $\triangle POD$
 $\triangle OJP$ $\triangle OEP$ $\triangle OEQ$ $\triangle OQJ$ $\triangle OBX$ $\triangle ODX$
 $\triangle OZC$ $\triangle ODZ$ $\triangle OYC$ $\triangle OYA$ $\triangle OVA$ $\triangle OBV$
 $\triangle AMP$ $\triangle ADP$ $\triangle COJ$ $\triangle CRJ$ $\triangle BNE$ $\triangle BSE$
 $\triangle DSQ$ $\triangle DLQ$ $\triangle EPJ$ $\triangle EJQ$ $\triangle QPE$ $\triangle PQJ$

10. (a);



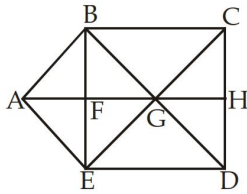
$\triangle ABC$ $\triangle AFE$ $\triangle GFO$ $\triangle GBH$ $\triangle GOH$ $\triangle HOI$
 $\triangle IOD$ $\triangle IDC$ $\triangle EOD$ $\triangle FEO$ $\triangle AGD$ $\triangle HEC$
 $\triangle BFI$

11. (b);



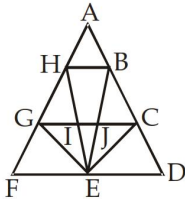
ADE, BDF, DEF and EFC $\rightarrow$ 4 Δ 's
 ABC $\rightarrow$ 1 Δ
 Total = 4 + 1 = 5 Δ 's

12. (c);



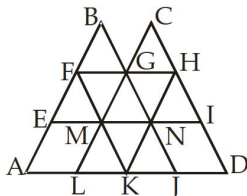
ABF, BFG, BCG, CGH, GHD, GED, EFG and AFE
 $\rightarrow$ 8 Δ 's
 ABG, BGE, AGE, ABE and GCD $\rightarrow$ 5 Δ 's
 BCD, CDE, BED and BCE $\rightarrow$ 4 Δ 's
 Total = 8 + 5 + 4 = 17 Δ 's

13. (b);



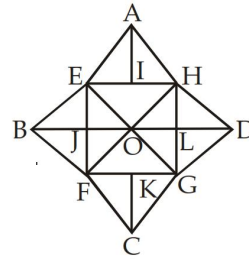
AHB, GHI, BJC, GFE, GIE, IJE, CEJ and CDE $\rightarrow$ 8 Δ 's
 HEG, BEC, HBE, JGE and ICE $\rightarrow$ 5 Δ 's
 FHE, GCE and BED $\rightarrow$ 3 Δ 's
 AGC $\rightarrow$ 1 Δ
 AFD $\rightarrow$ 1 Δ
 Total = 8 + 5 + 3 + 1 + 1 = 18 Δ 's

14. (b);



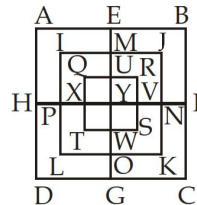
BFG, CGH, EFM, FMG, GMN, GHN, HNI, LMK,
 MNK and KNJ $\rightarrow$ 10 Δ 's
 FAK and KHD $\rightarrow$ 2 Δ 's
 BEN, CMI, GLJ and FHK $\rightarrow$ 4 Δ 's
 BAJ and CLD $\rightarrow$ 2 Δ 's
 Total = 10 + 2 + 4 + 2 = 18 Δ 's

15. (c);



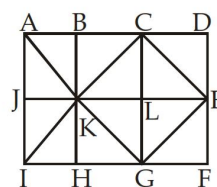
AEI, AIH, BEJ, BJE, CFK, CKG, DGL, DLH, EOJ,
 FOJ, FOG, LOG, HOL and HOE $\rightarrow$ 14 Δ 's
 EAH, FBE, BEO, EOF, BFO, FCG, GDH, HOD,
 HOG, GOD $\rightarrow$ 10 Δ 's
 EFH, EHG, FGH, EFG $\rightarrow$ 4 Δ 's
 Total = 14 + 10 + 4 = 28 Δ 's

16. (c);



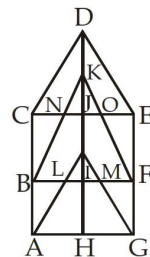
QUYX, URVY, YVSW and XYWT $\rightarrow$ 4 square's
 IMYP, MJNY, YNKO and PYOL $\rightarrow$ 4 square's
 AEYH, EBFY, YFCG, HYGD $\rightarrow$ 4 square's
 IJKL $\rightarrow$ 1 square
 QRST $\rightarrow$ 1 square
 ABCD $\rightarrow$ 1 square
 $\therefore$ Total number of square in the given figure = 4 +
 + 4 + 4 + 1 + 1 + 1 = 15 square

17. (c);



ABKJ, BCLK, CDEL, LEFG, KLGH and JKHI $\rightarrow$ 6
 squares
 CEGK $\rightarrow$ 1 square
 ACGI and BDFH $\rightarrow$ 2 squares
 $\therefore$ Total = 6 + 1 + 2 = 9 squares

18. (a);



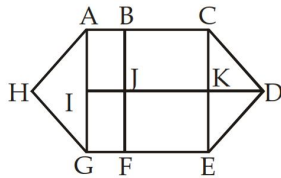
KJN, KJO, CNB, OEF, JIL, JIM, BLA and MFG $\rightarrow$

8 Δ 's
 CDJ, EDJ, NKO, JLM, JAH and JGH $\rightarrow$ 6 Δ 's
 BKI, FKI, CJA and EJG $\rightarrow$ 4 Δ 's
 CDE and AJG $\rightarrow$ 2 Δ 's
 BKF $\rightarrow$ 1 Δ
 Total = 8 + 6 + 4 + 2 + 1 = 21 Δ 's

Parallelograms :

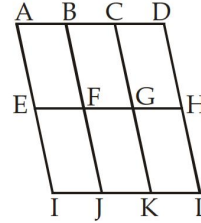
NJLB and JOFM $\rightarrow$ 2 ||gms
 CDKB, DEFK, BIHA and IFGH $\rightarrow$ 4 ||gms
 BKJA, KFGJ, CJIB, JEFI $\rightarrow$ 4 ||gms
 BFGA $\rightarrow$ 1 ||gm
 CDJA, DEGJ, CJHA, JEGH $\rightarrow$ 4 ||gms
 CEFB $\rightarrow$ 1 ||gm
 CEGA $\rightarrow$ 1 ||gm
 Total = 2 + 4 + 4 + 1 + 4 + 1 + 1 = 17 ||gms
 (Here not that the squares and rectangles are also counted among the parallelograms).

19. (b);



ABJI, BCKJ, IJFG and JKEF $\rightarrow$ 4
 ACKI, BCEF, IKEG and ABFG $\rightarrow$ 4
 ACEG $\rightarrow$ 1
 Total = 4 + 4 + 1 = 9 rectangles in the given figure.

20. (b);



ABFE, BCGF, CDHG, EFJI, FGKJ and GHLK. $\rightarrow$ 6 ||gms
 ACEG, BDHF, EGKI, FHLJ, ABJI, BCKJ and CDLK $\rightarrow$ 7 ||gms
 ADHE and EHLI $\rightarrow$ 2 ||gms
 ACKI, BDLJ $\rightarrow$ 2 ||gms
 ADLI $\rightarrow$ 1 ||gm
 Total = 6 + 7 + 2 + 2 + 1 = 18 parallelograms in the figure.